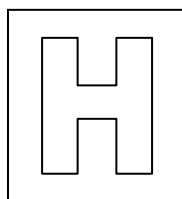


Candidate Name: _____

Class Adm No

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2011 Preliminary Examination II

Pre-university 3

H2 Biology

9648/03

19 September 2011

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

For Examiner's Use	
1	
2	
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5	
Total	

This question paper consists of 16 printed pages.

[Turn over

Answer **all** questions.

1. Huntington's disease (HD) results from the inheritance of an autosomal dominant allele.

- (a) Genetic screening for HD can be carried out, using a process similar to genetic fingerprinting, to find the length of a repeated triplet, or 'stutter', in an allele.

After treatment with a restriction enzyme, fragments of DNA of different lengths are separated by gel electrophoresis.

- (i) Describe the role of a restriction enzyme in this technique.

..... cuts DNA into, pieces / fragments / AW ;
 at specific sites ;
 close to, but not in stutter ;
 detail of site ; [4-6 base pairs, palindromic]

 [3]

- (ii) Explain why fragments of DNA of different lengths can be separated by gel electrophoresis.

..... negatively charged due to phosphate group;
 attracted to anode ;
 smaller fragments travel further (towards anode) / ora ;
 smaller fragments have less impedance / AW / ora ;

 [3]

- (b) Fig. 1.1 shows a family's history of HD.

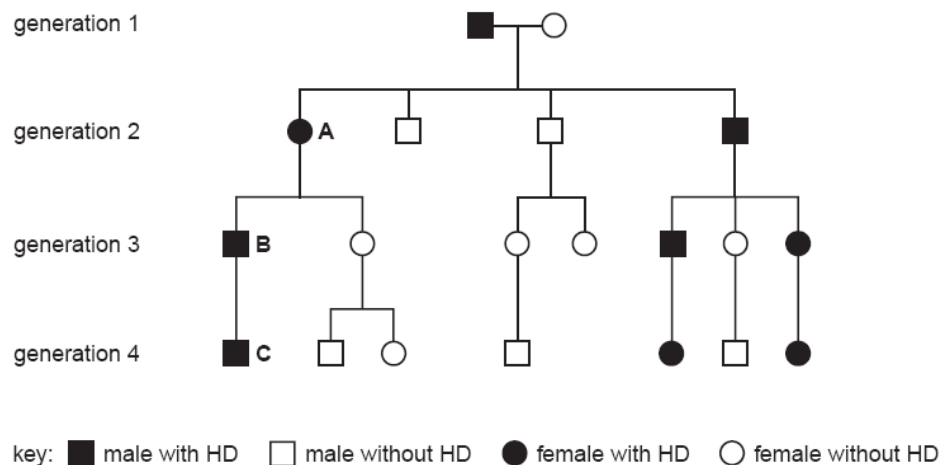


Fig. 1.1

DNA from individuals **A**, **B** and **C** from the family shown in Fig. 1.1 was analysed. The resulting banding patterns are shown in Fig. 1.2.

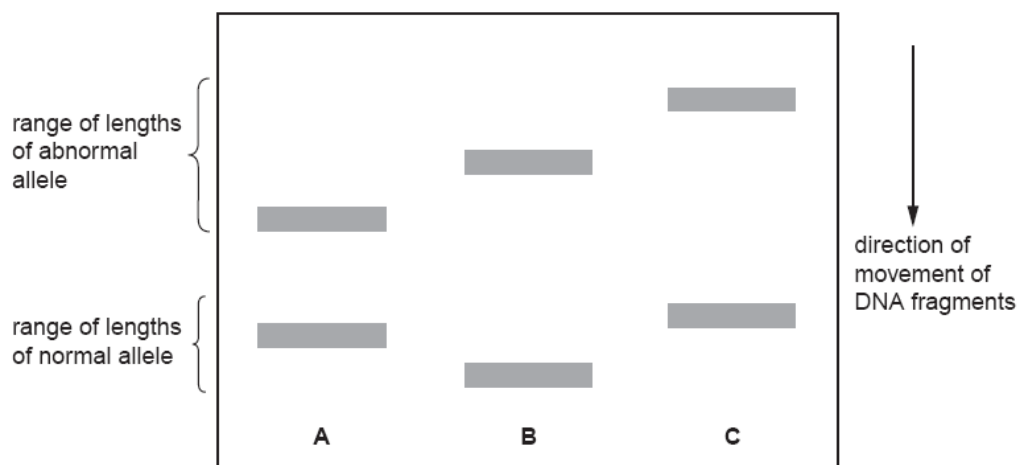


Fig. 1.2

Suggest why the DNA of the following bands in Fig. 1.2 are not the same length:

(i) the three normal alleles

..... inherited from three different individuals/ each parent (unaffected
parents of A, B and C) ; [1]

(ii) the three abnormal alleles

length of stutter increases in each generation/ C longer than A/
different sizes of stutter ; [1]

Down's syndrome in humans usually results from acquiring an extra chromosome 21 from the mother.

A small section of chromosome 21, the so-called Down's syndrome critical region (DSCR) causes the abnormalities associated with the syndrome.

Precise regulation of gene expression by a particular group of transcription factors is essential for normal development. In inactive cells the transcription factors are found in the cytoplasm, but in active cells they move into the nucleus.

In humans, a gene, D, has been identified in the DSCR that affects a transcription factor. Mice that have been genetically engineered to over-express D show many of the symptoms of Down's syndrome.

The nuclear / cytoplasmic ratio of the transcription factors and the expression of the genes that they control was measured in the genetically engineered mice and in normal mice. The results are shown in Table 1.3.

	normal mice	mice genetically engineered to over-express D
nuclear / cytoplasmic ratio of transcription factors	2.75	0.75
relative expression of genes controlled by these transcription factors	×12	×1

Table 1.3

(c) With reference to the information given in Table 1.3 and information above,

(i) describe the effect of the over-expression of the human gene, D, in mice;

...	nuclear:cytoplasmic ratio of transcription factors higher in normal / ora ;	
...	A fewer transcription factors in nucleus / more in cytoplasm	
...	A use of figures to show effect	
...	more genes (controlled by these transcription factors) expressed in normal mice ;	
...	A less transcription / use of figures / ora	 [2]

(ii) suggest why the genetically engineered mice show many of the symptoms of Down's syndrome.

Down's syndrome result of, trisomy / 3 chromosomes 21 ;	
thus x1.5 normal expression ;	
over-expression of D equivalent to, extra chromosome 21 / (Down's) symptoms ;	
(essential) genes not transcribed ; A 'not regulated' / not switched on' R DSCR	
transcription factors, not moving to nucleus / staying in cytoplasm ;	
cells remain inactive ;	 [3]
so normal, growth / development, does not occur ;	

- (d) Suggest what benefits and hazards arise from genetically engineering mice to express human genes.

Benefits

- benefits
 idea that provides animal model for research / **AW** ;
 allows, drugs / treatment, to be tested without risking humans ;
 **AVP** ; e.g. no ethical problems as with using humans

- hazards
 findings in animals may not relate to humans ;
Haz risk of transgene, entering germ line / escaping from containment ;
 unforeseen effect on mice ;
 escapes and breeds with, wild population / non GM mice / GM mice for another condition ;
 **AVP** ; e.g. use of animals for research unethical / ref to animal rights

..... [4]

[Total: 17]

2. One alternative to the use of embryonic stem cells in human therapy are recently-developed methods of genetically reprogramming the nuclei of differentiated adult cells so that they regain the pluripotency of embryonic stem (ES) cells.

An induced pluripotent stem cell, or IPS cell, is a stem cell that has been created from an adult cell such as a skin, liver, stomach or other mature cell through the introduction of genes that reprogram the cell and transform it into a cell that has all the characteristics of an embryonic stem cell.

So far, IPS cells appear to exhibit the same key features of ES cells: the ability to differentiate from a blank-slate state to any of the 220 types of cells in the human body, and the ability to reproduce indefinitely in culture.

- (a) Stem cells can be divided into four main types: adult stem cells, embryonic stem cells, fetal stem cells and umbilical stem cells.

- (i) Identify a limited Adult stem cells; e into

..... [1]

- (ii) Explain how will the potency of the stem cell you have identified in (i) differs from IPS cells.

.....	Adult stem cells – multipotent	
.....	IPS – pluripotent;	
.....	Adult stem cells – limited range of tissue	
.....	IPS – all organs in an organisms but not the whole organism;	

..... [2]

- (b) Suggest why a bone marrow cell cannot be induced to produce a skin cell and a ~~skin cell cannot be induced to produce a bone marrow cell~~

..... Genes not required for a particular cell line are methylated;
signal is required in order to switch on or off certain genes;

..... [1]

- (c) Besides the problem of availability of human ES cells, explain why human IPS cells are good substitute to human ES cells.

..... Avoid destroying a human life;

..... [1]

Both stem cells and cancer cells have the ability to proliferate

(d) Identify two ways which stem cells and cancer cells differ:

Stem cells are known to divide asymmetrically, with one daughter cell remaining a largely quiescent stem cell and the other undergoing further rapid divisions as the progeny differentiate.

Cancer cells divide symmetrically.

Divide when necessary

Uncontrolled cell division

[2]

(e) State one normal role of stem cells in living organism.

.....
regenerate;
repair; (R: repair cells)
growth;
.....

[1]

[Total: 8]

3. Many Cowpeas, *Vigna unguiculata*, are leguminous plants which are used as forage crops. The plants contain an inhibitor of the enzyme trypsin. Trypsin is a protease that is found in the digestive tracts of many animals including insects. This provides the plant with resistance against various insect pests.

The cowpea trypsin inhibitor (CpTi) is a small polypeptide of 80 amino acids. The allele coding for its production has been successfully transferred to other crops, such as potato and tobacco, to increase their yield.

- (a) Leaf cells from tobacco plants were genetically engineered to produce CpTi by means of a plasmid from the bacterium, *Agrobacterium tumefaciens*. The plasmid included:

- the coding sequence for CpTi
- a regulatory sequence known to work in plants
- a sequence giving resistance to kanamycin, an antibiotic which kills both bacterial cells and plant cells.

- (i) Describe two properties of bacterial plasmids that allow them to be used as cloning vectors.

1	<u>one origin of replication;</u>	
.....	small size; (OWTTE)	
2	A: selectable marker, usually antibiotic resistance (stated in qns)	
.....	A: regulatory sequence (stated in qns)	
.....	A: Multiple cloning site (commercial plasmids)	 [2]

- (ii) Explain the importance of the presence, in the plasmid, of the sequence giving resistance to kanamycin.

.....	selectable marker, used to identify cells that contain transformed plasmids.;	
.....	survival of transformed cells;	
.....	death of non-transformed cells;	
.....		
.....		
.....		[3]

- (b) The successfully genetically engineered leaf cells were grown into plantlets using tissue culture.

Outline a procedure for growing plantlets from cells in tissue culture.

.....

.....

.....

.....

.....

.....

.....

- ... explant cells cultured in growth/nutrient medium (cytokinin);
stimulates mitosis/cell division to form callus (A: explant divides to form callus);
... subculturing of callus (into another medium);
addition of identified ratio of other growth regulators (auxin and gibberellin);
callus differentiate/grow into plantlet; [5]

No details of obtaining the cells were required, nor was the treatment of the plantlets once they had been formed.

The effectiveness of CpTi against insect pests was tested using two groups of tobacco plants.

- Plants genetically engineered to express the allele coding for CpTi.
- Control plants which had been genetically engineered by inserting the allele coding for CpTi in the wrong orientation. These plants did not produce CpTi.

Equal numbers of each type of plant were infested with equal numbers of newly hatched larvae of an insect pest and left for seven days.

The mean percentage of leaf area eaten and the mean mass of surviving insects were then found. The results are shown in Fig. 3.1.

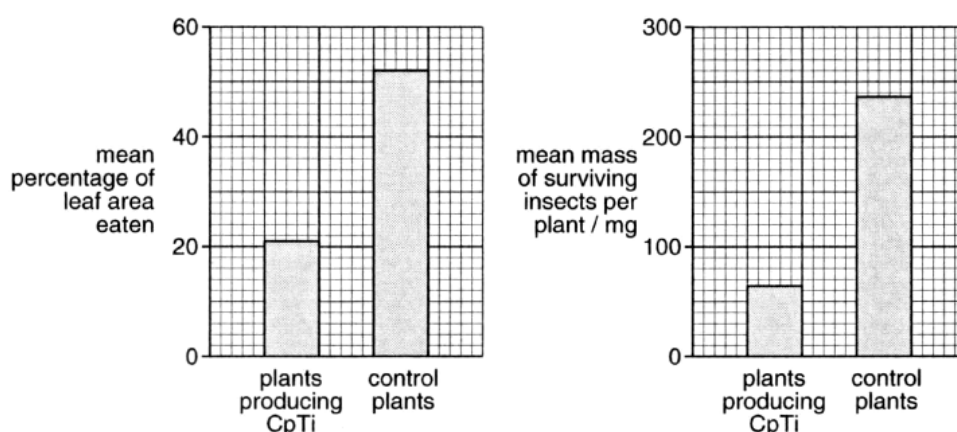


Fig. 3.1

With reference to Fig. 3.1,

(c)

(i) describe the effect of CpTi on yield of tobacco leaves

- Lower area of leaf eaten;
52% in control, 21% in plants producing CpTi;
- accurate figures relating to the mean percentage of leaf area eaten required
relating to the mass of surviving insects per plant were not relevant*
- [2]

(ii) suggest an explanation of the effect of the trypsin inhibitor, CpTi, on insects.

- refer to the lack of digestion of protein; (R: rather than 'food')
- Enzyme inhibition, hence active sites not able to bind to substrate;
- insects would die of malnutrition or starvation; Ref: accumulation of undigested food in the gut or indigestion.
- [3]

[Total: 15]

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4. Transformation of plasmid DNA into *E. coli* using the heat shock method is a basic technique of molecular biology. It consists of inserting a foreign plasmid or ligation product into bacteria.

After a short incubation in ice, a mixture of chemically competent bacteria and DNA is placed at a warm temperature for 45 seconds (heat shock) and then placed back in ice. This traditional protocol can be used successfully to transform most commercially available competent bacteria. Fig. 4.1 shows a simple illustration of bacteria transformation through heat shock.

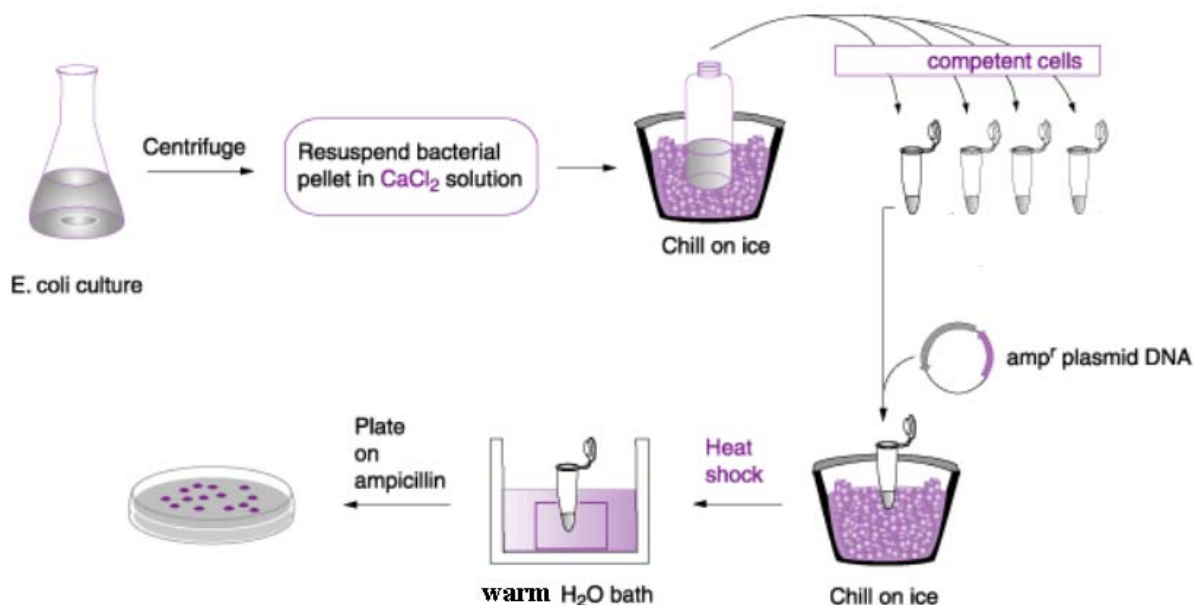


Fig. 4.1

You are to design an experiment to investigate the effect of varying the heat-shock temperature on the transformation of the bacterium *Escherichia coli* using a plasmid vector.

[Total: 12]

- Your plan should have a clear and helpful structure to include:
- a description of the method used including the scientific reasoning behind the method,
- an explanation of the dependent and independent variables involved,
- relevant, clearly labelled diagrams,
- how you will record your results and ensure they are as accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms,
- relevant risks and precautions taken.

Independent Variable:

1. ref. to water baths at different temperature
2. at least 5 different temperatures
3. ref. to suitable range (any range within 0 – 70°C)
4. ref to retesting within approximate optimum zone

Dependent Variable:

5. ref. to highest bacteria growth rate is optimum

Control Variables:

6. includes 3 variables to be controlled - incubation temp, time, same batch of nutrient medium used, same plasmid vector with same gene content, same E.coli strain

Procedure:

7. gives details on the aseptic preparation of control and experimental plates;
8. details on measuring the results/outcomes of the different heat shock temperatures on bacterial transformation with details on the use of genetic markers;

Data processing:

9. proposed layout of results tables and graphs with clear headings and labels,
10. computation of growth rate of bacteria
11. due regard to producing reliable results (e.g. repeats experiment with replicates at least twice with reference to appropriate statistical analysis

Safety:

12. safety procedures are expanded with reasons e.g. use of antiseptic solutions to prevent growth of harmful microorganisms

[illegible]

SCID

- S1 more males affected (than females);
 S2 recessive;
 S3 (most common form is) sex-linked (X chromosome);
 S4 mutation of IL2RG (interleukin-2 receptor gamma) gene (for gamma chain on lymphocyte receptor);
 S5 ADA (adenosine deaminase) (deficiency);
 S6 mutation of gene on chromosome 20;
 S7 prevents formation of, lymphocytes / T cells / B cells;
 S8 toxic metabolites not broken down so lymphocytes killed;
 S9 so susceptible to opportunistic infections;
 S10 children usually die (if not treated);
 S11 pneumonia / PCP / meningitis / chickenpox / measles / other valid;

- 1 gene therapy is a technique for introducing 'normal', allele / gene;
 2 new, allele / gene, generates, functional/correct, protein product;
 3 restores target cell to normal state / corrects function of cell / corrects phenotype;

viral delivery systems

- 4 viruses are genetically engineered to carry 'normal' human gene;
 5 adenoviruses / retroviruses / MMLV (for SCID);
 6 virus releases genetic material into target cells;
 7 in SCID the gene is inserted (by virus) into, bone marrow cells / blood stem cells / lymphocytes, ex vivo / in vitro;
 8 in CF the gene is inserted into cells within body;

non-viral delivery systems

- 9 direct introduction of DNA into target cells;
 (gene gun or other direct method e.g. micropipette)
 10 creation of, lipid sphere / liposome, which carries DNA;
 11 Liposome can / use of electroporation to, pass through target cell membrane;
 12 link DNA to molecule that will bind to receptors in cell membrane;
 13 DNA is engulfed by membrane;
 14 possibility of introducing 47th chromosome into cells;

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5. With reference to genetic diseases,

(a) Describe SCID (severe combined immunodeficiency) [8]

(b) Explain gene therapy treatment, using both viral and non-viral delivery systems. [6]

(a) Discuss the factors which prevent gene therapy from becoming an effective treatment. [6]

[Total: 20]

1. difficult to get DNA to integrate into the target cell genome;
2. results can be short-lived / patients need to be treated on a frequent basis;
3. ref. problem with controlling the activity of gene expression;
4. risk of stimulating immune response;
5. virus vector may, regain / develop, virulence;
6. incorrect insertion of gene may cause, named / described, problem to patient;
7. possible problem with finding vector for large gene;
8. many genetic diseases are a result of presence of many defective genes;
9. impossible to introduce many genes at same time;
10. problem dealing with, dominant condition / non dividing cells;
11. ref. problem with liposomes; (toxicity / low efficiency)